

Homework 18

Please read the following from Durrett:

- Section 6.1, just through the end of Example 1. Durrett focuses on independent random variables in this section.
- Section 6.2, through the end of Example 6.8. The formulas in this part do not require the random variables you are summing to be independent.

Please work the following exercises from Chapter 6:

1 (textbook answer: -1 (dollar). Use the methods of Chapter 6 to explain how to get this value without calculating the distribution of the bettor's total winnings.)

Let X_i be the random variable representing the bettor's net winnings on bet i , measured in dollars. The possible values of X_i are 1 (with probability $18/38$) and -1 (with probability $20/38$). We thus have $E[X_i] = \frac{18}{38} - \frac{20}{38} = -\frac{2}{38} = -\frac{1}{19}$.

The bettor's overall net winnings are given by the random variable $X = X_1 + X_2 + \cdots + X_{19}$. The expected winnings are thus $E[X] = E[X_1] + \cdots + E[X_{19}] = 19 \left(-\frac{1}{19} \right) = -1$.

If this kind of analysis of a gambling game feels unfamiliar, look back at similar examples from Chapter 1, both in the text and in homework.

2 (Explain using the method of indicator variables. Let N be the random variable representing the number aces in a random hand of 13 cards drawn from a standard deck. Represent N as a sum of indicator variables and use the additivity of expected value to calculate $E[N]$.)

We can solve this question in several ways, as with the class activity about counting hearts in a hand of five cards.

First method: Let A_i ($i = 1, \dots, 13$) be the event that the i th card drawn is an ace. We have $P(A_i) = \frac{4}{52} = \frac{1}{13}$. (In position i of the shuffled deck, we are equally likely to find any of the 52 cards. Of the 52, 4 are aces, and so the probability of an ace in position i is $4/52$.)

Let X_i be the indicator of the event A_i . As we discussed in class, $E[X_i] = P(A_i) = \frac{1}{13}$. We have $N = X_1 + \cdots + X_{13}$. Therefore, $E[N] = E[X_1] + \cdots + E[X_{13}] = \frac{13}{13} = 1$.

Second method: Let B_j ($j = 1, 2, 3, 4$) be the event that the j th ace is one of the 13 cards we draw. In other words, B_j is the event that the j th ace is shuffled into one of the top 13 positions (out of 52) in the deck. Therefore, $P(B_j) = \frac{13}{52}$.

Let Y_j be the indicator of the event B_j . We then have $E[Y_j] = \frac{13}{52}$. We can write N as $N = Y_1 + Y_2 + Y_3 + Y_4$, and so $E[N] = E[Y_1] + E[Y_2] + E[Y_3] + E[Y_4] = \frac{13}{52} + \frac{13}{52} + \frac{13}{52} + \frac{13}{52} = 4 \left(\frac{13}{52} \right) = \frac{52}{52} = 1$.

3 (textbook answer: $-3/5$. Use the methods of Chapter 6 to explain how to get this value. Write X and Y as sums of indicator random variables and use the additivity of the expected value to calculate $E[X - Y]$.)

As with the previous question, we have two solution methods.

First method: Let A_i ($i = 1, \dots, 10$) be the event that boy i is picked. We have $P(A_i) = \frac{3}{25}$. There are many ways to calculate that probability.

- There are $C_{25,3} = \frac{(25)(24)(23)}{3!}$ equally likely ways to pick three students from the class of 25. There are $C_{24,2} = \frac{(24)(23)}{2!}$ ways to pick three students, one of whom is boy i . We thus have $P(A_i) = \frac{C_{24,2}}{C_{25,3}} = \frac{3}{25}$.
- Imagine picking the students by arranging them in a random order and taking the first three. There are 25 equally likely positions for boy i . Of those 25, 3 positions lead to picking boy i . Therefore, $P(A_i) = \frac{3}{25}$.

Let U_i be the indicator for A_i . As before, we have $E[U_i] = \frac{3}{25}$. We can write X as $X = U_1 + \dots + U_{10}$, and so $E[X] = \frac{30}{25} = \frac{6}{5}$.

Let B_j ($j = 1, \dots, 15$) be the event that girl j is picked. As above, we have $P(B_j) = \frac{3}{25}$. Let V_j be the indicator for B_j , so that $E[V_j] = \frac{3}{25}$. Writing $Y = V_1 + \dots + V_{15}$, we find $E[Y] = \frac{45}{25} = \frac{9}{5}$.

We now have $E[X - Y] = E[X] - E[Y] = \frac{6}{5} - \frac{9}{5} = -\frac{3}{5}$.

Second method: Let C_i ($i = 1, 2, 3$) be the event that the i th student picked is a boy. We have $P(C_i) = \frac{10}{25}$. Letting W_i be the indicator of C_i , we find $X = W_1 + W_2 + W_3$, and so $E[X] = \frac{10}{25} + \frac{10}{25} + \frac{10}{25} = \frac{6}{5}$ (as before).

Let D_i ($i = 1, 2, 3$) be the event that the i th student picked is a girl. We have $P(D_i) = \frac{15}{25}$. Letting Z_i be the indicator of D_i , we find $Y = Z_1 + Z_2 + Z_3$, and so $E[Y] = E[Z_1] + E[Z_2] + E[Z_3] = \frac{15}{25} + \frac{15}{25} + \frac{15}{25} = \frac{9}{5}$ (as before).

We thus again get $E[X - Y] = \frac{6}{5} - \frac{9}{5} = -\frac{3}{5}$.

Please also answer the following two questions. You started one of them on a class worksheet.

1. A few weeks ago, we modeled gaps (measured in hours) between e-mail messages that I receive as random variables with the exponential(5) distribution. Use this model to calculate how long I should expect it to take for me to get 10 e-mail messages. Reason this out using additivity of expected values, explaining your thinking clearly.

Let X_i be the waiting time for the i th e-mail message. That means X_1 is the waiting time for the first e-mail to arrive, X_2 is the gap between the first and second, and so forth. The total waiting time Y for the 10 messages is $X_1 + \dots + X_{10}$. Since $E[X_i] = \frac{1}{5}$, we have $E[Y] = 10 \left(\frac{1}{5} \right) = 2$. The units of measurement are hours.

2. Imagine the chance experiment of drawing a hand of five cards from a standard deck. Let N be the random variable that counts the number of suits in your hand. This variable takes the values 1, 2, 3, 4. For example, if you have 2 hearts, 1 diamond, and 2 clubs in your hand, then N for this outcome is 3.
- a. Number the suits 1, 2, 3, 4 (in any way). Let A_i be the event that your hand has at least one card of suit i . Calculate $P(A_i)$.

$$P(A_i) = 1 - \frac{C_{39,5}}{C_{52,5}}$$

The complement A_i^c is the event that the hand has no cards of suit i . We can think of calculating $P(A_i^c)$ as an urn problem. We count the number of 5-card hands with no cards from suit i ($C_{39,5}$) and the number of 5-card hands with no restrictions ($C_{52,5}$). We thus find $P(A_i^c) = \frac{C_{39,5}}{C_{52,5}}$, and so $P(A_i) = 1 - \frac{C_{39,5}}{C_{52,5}}$.

- b. Use the indicator variables for the events A_i to calculate $E[N]$. (You should get roughly 3.114.) Your explanation should include the indicator variables that you use and their expected values.

Let X_i be the indicator for the event A_i . We have $N = X_1 + X_2 + X_3 + X_4$, and so $E[N] = E[X_1] + E[X_2] + E[X_3] + E[X_4] = 4 \left(1 - \frac{C_{39,5}}{C_{52,5}} \right) \approx 3.1138655462$.